Tutorial 7

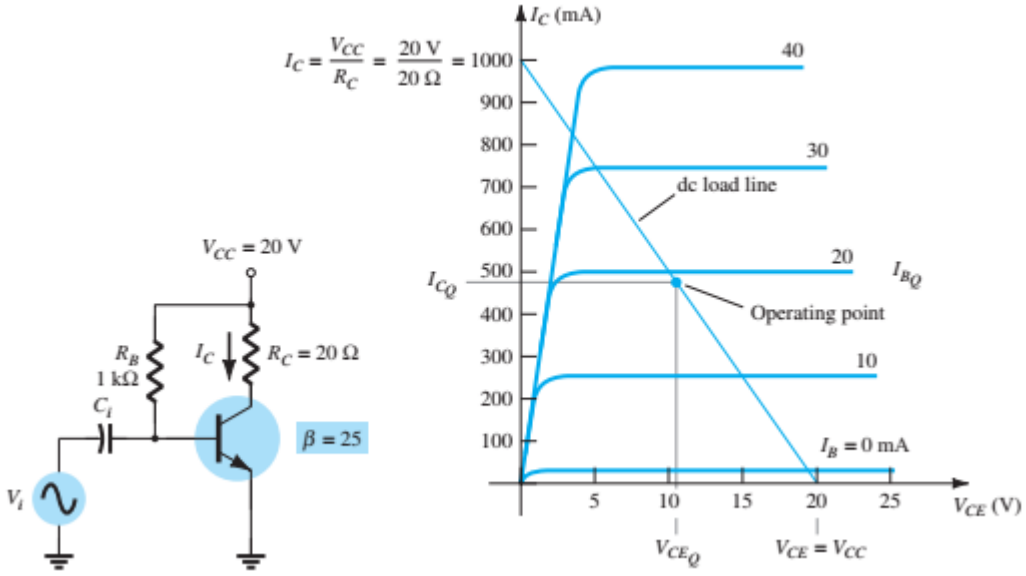
Question 1: What is a power amplifier? Discuss the differences between a power and a voltage amplifier.

Question 2: Explain the terms related to a power amplifier: Collector efficiency, power dissipation capability, heat sink, Distortion and thermal runaway.

Question 3: Based on the cycle of operation, how can you classify power amplifiers?

Problem 1:

Calculate the input power, output power, and efficiency of the amplifier circuit shown below for an input voltage that results in a base current of 10 mA peak.



Problem 2:

For a class B amplifier providing a 20-V peak signal to a 16-ohm load (speaker) and a power supply of VCC = 30 V, determine the input power, output power, and circuit efficiency.